Scenario

In this scenario, you're a security analyst investigating traffic to a website.

You'll analyze a network packet capture file that contains traffic data related to a user connecting to an internet site. The ability to filter network traffic using packet sniffers to gather relevant information is an essential skill as a security analyst.

You must filter the data in order to:

- identify the source and destination IP addresses involved in this web browsing session,
- examine the protocols that are used when the user makes the connection to the website, and
- analyze some of the data packets to identify the type of information sent and received by the systems that connect to each other when the network data is captured.

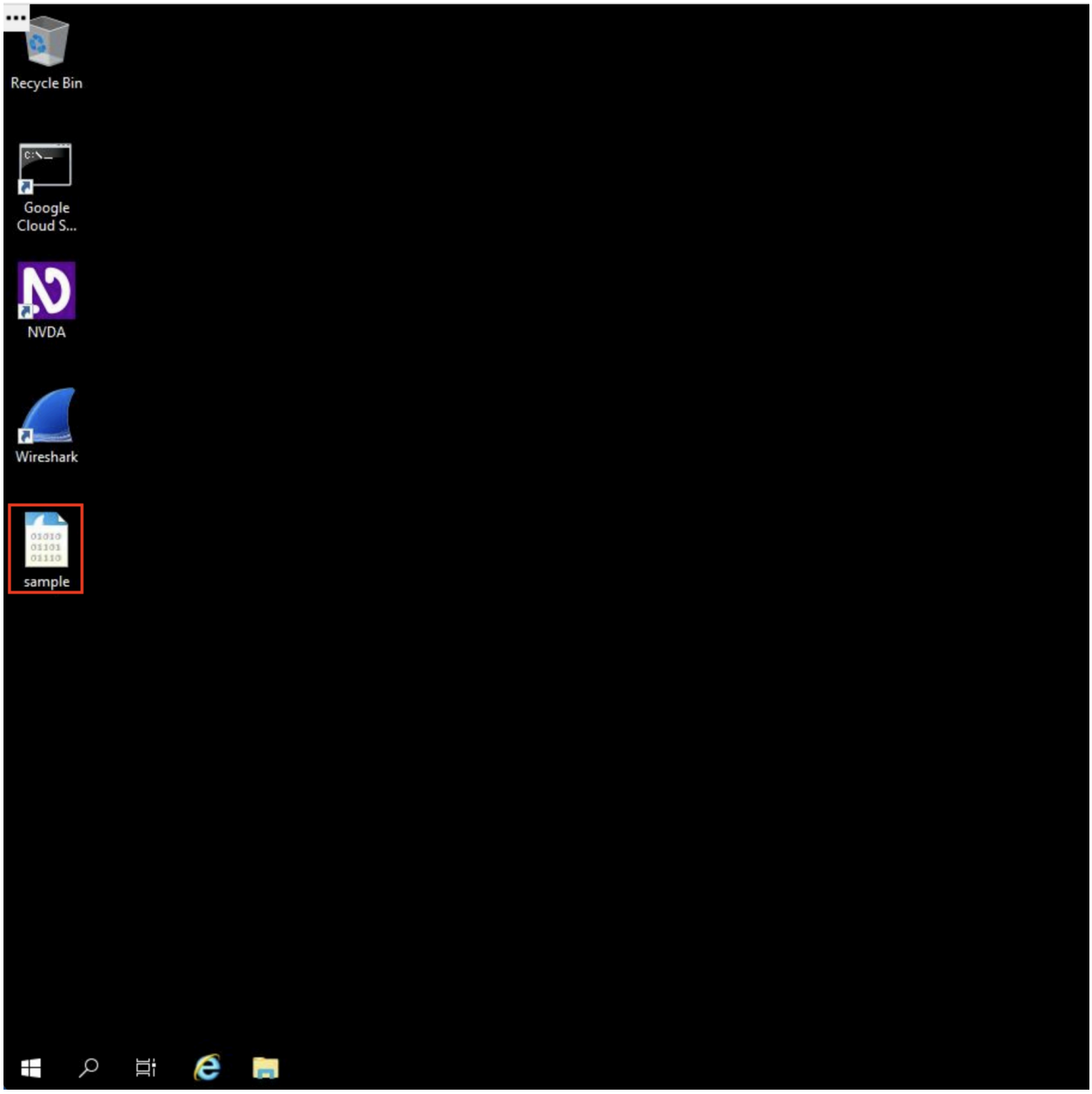
Here's how you'll do this: **First**, you'll open the packet capture file and explore the basic Wireshark graphic user interface. **Second**, you'll open a detailed view of a single packet and explore how to examine the various protocol and data layers inside a network packet. **Third**, you'll apply filters to select and inspect packets based on specific criteria. **Fourth**, you'll filter and inspect UDP DNS traffic to examine protocol data. **Finally**, you'll apply filters to TCP packet data to search for specific payload text data.

You're ready to use Wireshark to inspect network packet data!

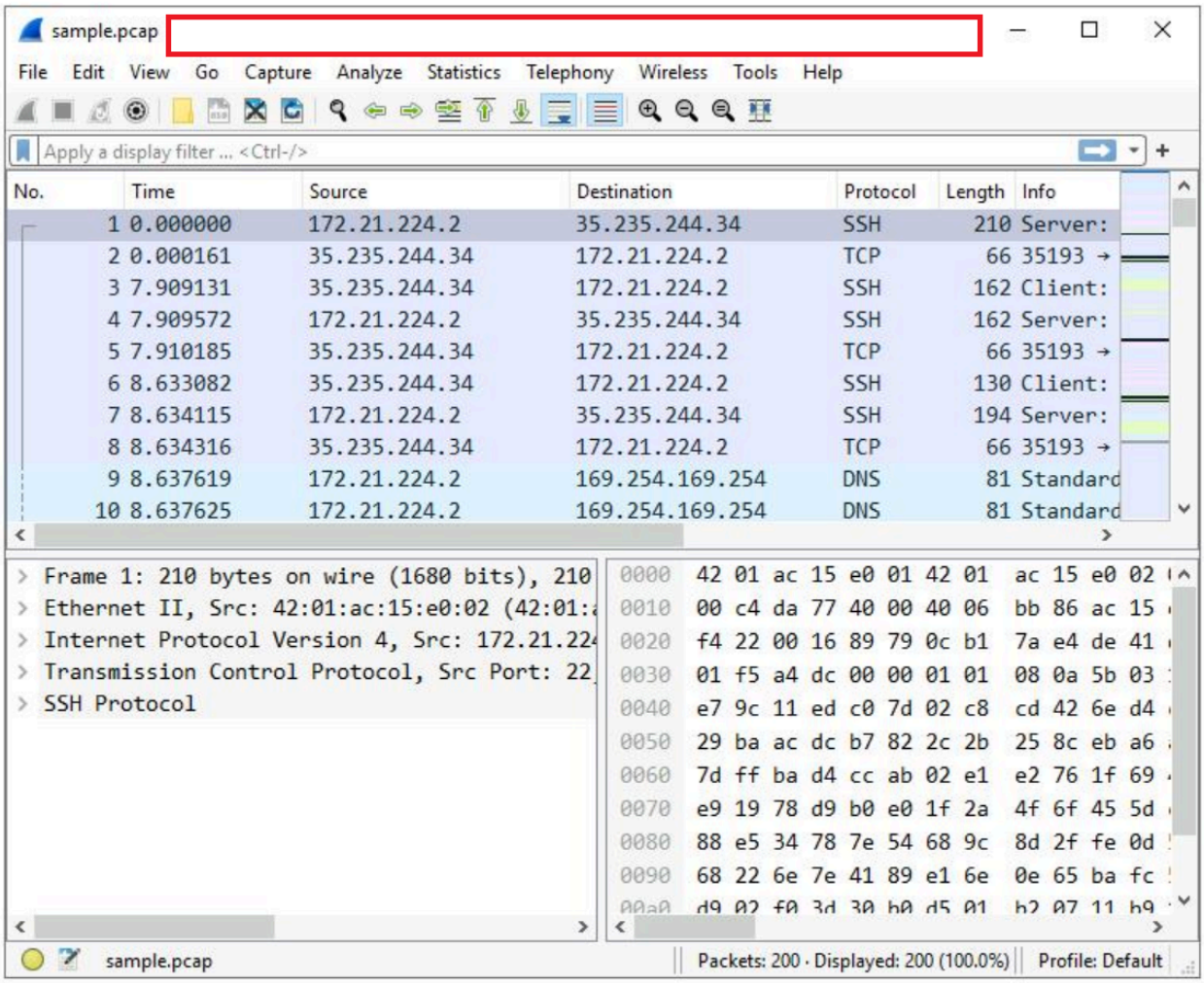
Task 1. Explore data with Wireshark

In this task, I opened a network packet capture file that contains data captured from a system that made web requests to a site. You need to open this data with Wireshark to get an overview of how the data is presented in the application.

1. To open the packet capture file, double-click the **sample** file on the Windows desktop. This will start Wireshark.



2. Double-click the Wireshark title bar next to the **sample.pcap** filename to maximize the Wireshark application window.



For now, here is an overview of the key property columns listed for each packet:

- **No.** : The index number of the packet in this packet capture file
- **Time**: The timestamp of the packet
- **Source**: The source IP address
- **Destination**: The destination IP address
- **Protocol**: The protocol contained in the packet
- **Length**: The total length of the packet
- **Info**: Some information about the data in the packet (the payload) as interpreted by Wireshark

Not all the data packets are the same color. Coloring rules are used to provide high-level visual cues to help you quickly classify the different types of data. Since network packet capture files can contain large amounts of data, you can use coloring rules to quickly identify the data that is relevant to you. The example packet lists a group of light blue packets that all contain DNS traffic, followed by green packets that contain a mixture of TCP and HTTP protocol traffic.

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Task 2. Apply a basic Wireshark filter and inspect a packet

In this task, I open a packet in Wireshark for more detailed exploration and filter the data to inspect the network layers and protocols contained in the packet.

1. Enter the following filter for traffic associated with a specific IP address. Enter this into the **Apply a display filter...** text box immediately above the list of packets:

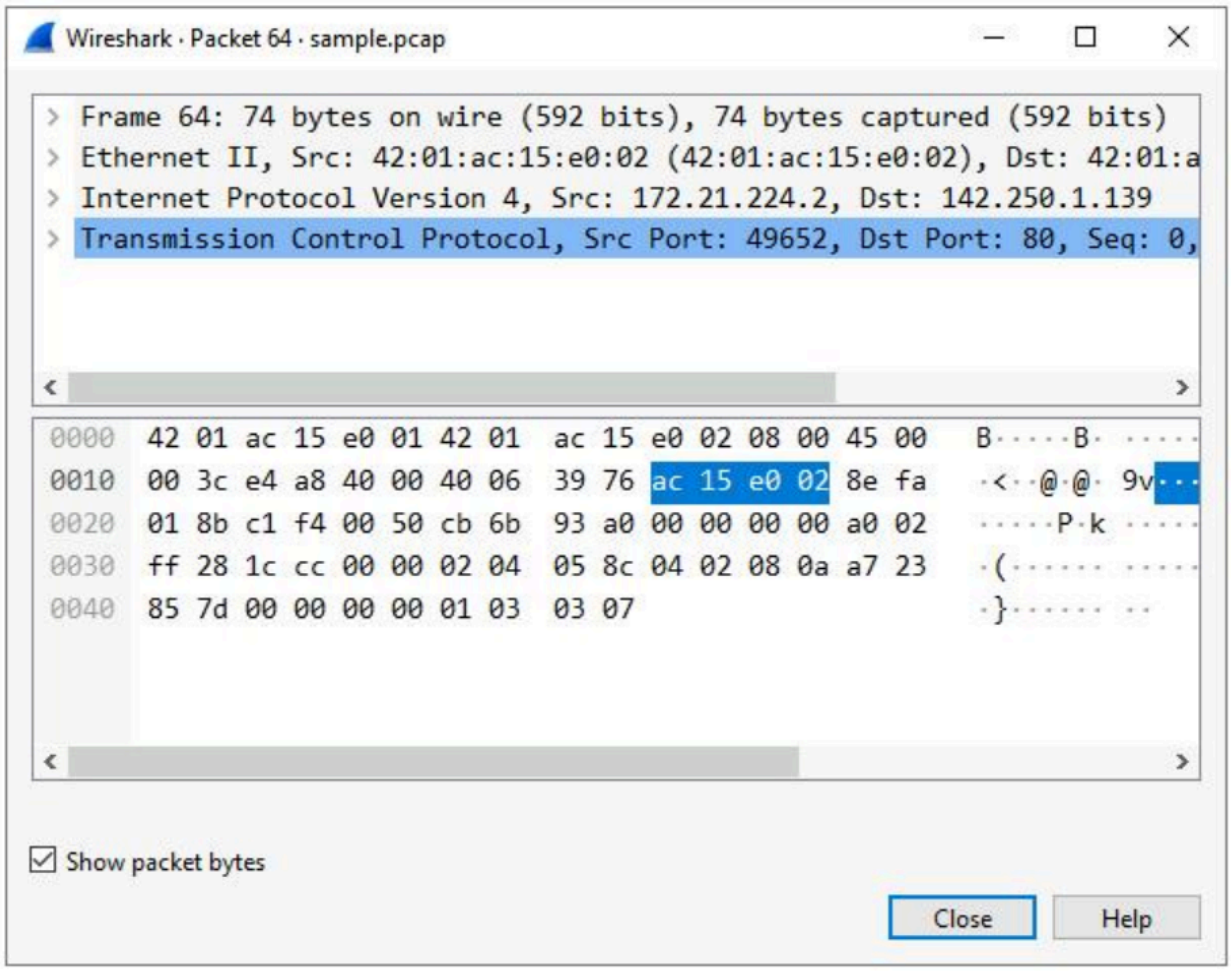
   ip.addr == 142.250.1.139

2. Press **ENTER** or click the **Apply display filter** icon in the filter text box.

The list of packets displayed is now significantly reduced and contains only packets where either the source or the destination IP address matches the address you entered. Now only two packet colors are used: **light pink** for `ICMP` protocol packets and **light green** for `TCP` (and `HTTP`, which is a subset of `TCP`) packets.

3. Double-click the first packet that lists **TCP** as the protocol.

This opens a packet details pane window:



7. Double-click **Internet Protocol Version 4** again to collapse that subtree and then double-click the **Transmission Control Protocol** subtree.

This provides detailed information about the TCP packet, including the source and destination TCP ports, the TCP sequence numbers, and the TCP flags.

The source port and destination port listed here match the source and destination ports in the info column of the summary display for this packet in the list of all of the packets in the main Wireshark window

Task 3. Use filters to select packets

In this task, I used filters to analyze specific network packets based on where the packets came from or where they were sent to. You'll explore how to select packets using either their physical Ethernet Media Access Control (MAC) address or their Internet Protocol (IP) address.

1. Enter the following filter to select traffic for a specific source IP address only. Enter this into the **Apply a display filter...** text box immediately above the list of packets:

ip.src == 142.250.1.139

2. Press **ENTER** or click the **Apply display filter** icon in the filter text box.

A filtered list is returned with fewer entries than before. It contains only packets that came from **142.250.1.139**.

3. Click the **X Clear display filter** icon in the Wireshark filter bar to clear the IP address filter.
4. Enter the following filter to select traffic for a specific destination IP address only:

ip.dst == 142.250.1.139

5. Press **ENTER** or click the **Apply display filter** icon in the filter text box.

A filtered list is returned that contains only packets that were sent to **142.250.1.139**.

6. Click the **X Clear display filter** icon in the Wireshark filter bar to clear the IP address filter.
7. Enter the following filter to select traffic to or from a specific Ethernet MAC address. This filters traffic related to one MAC address, regardless of the other protocols involved:

   eth.addr == 42:01:ac:15:e0:02

8. Press **ENTER** or click the **Apply display filter** icon in the filter text box.
9. Double-click the first packet in the list. You may need to scroll back to display the first packet in the filtered list.
10. Double-click the **Ethernet II** subtree if it is not already open.

The MAC address you specified in the filter is listed as either the source or destination address in the expanded Ethernet II subtree.

11. Double-click the **Ethernet II** subtree to close it.
12. Double-click the **Internet Protocol Version 4** subtree to expand it and scroll down until the **Time to Live** and **Protocol** fields appear.

What is the protocol contained in the Internet Protocol Version 4 subtree from the first packet related to MAC address 42:01:ac:15:e0:02?

The **Protocol** field in the **Internet Protocol Version 4** subtree indicates which IP internal protocol is contained in the packet.

Task 4. Use filters to explore DNS packets

In this task, I used filters to select and examine DNS traffic. Once you‘ve selected sample DNS traffic, you’ll drill down into the protocol to examine how the DNS packet data contains both queries (names of internet sites that are being looked up) and answers (IP addresses that are being sent back by a DNS server when a name is successfully resolved).

1. Enter the following filter to select UDP port **53** traffic. DNS traffic uses UDP port **53**, so this will list traffic related to DNS queries and responses only. Enter this into the **Apply a display filter...** text box immediately above the list of packets:

udp.port == 53

2. Press **ENTER** or click the **Apply display filter** icon in the filter text box.
3. Double-click the first packet in the list to open the detailed packet window.
4. Scroll down and double-click the **Domain Name System (query)** subtree to expand it.
5. Scroll down and double-click **Queries**.

I noticed that the name of the website that was queried is **opensource.google.com**.

6. Click the **X** icon to close the detailed packet inspection window.
7. Double-click the fourth packet in the list to open the detailed packet window.
8. Scroll down and double-click the **Domain Name System (query)** subtree to expand it.
9. Scroll down and double-click **Answers**, which is in the **Domain Name System (query)** subtree.

The Answers data includes the name that was queried (**opensource.google.com**) and the addresses that are associated with that name.

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Task 5. Use filters to explore TCP packets

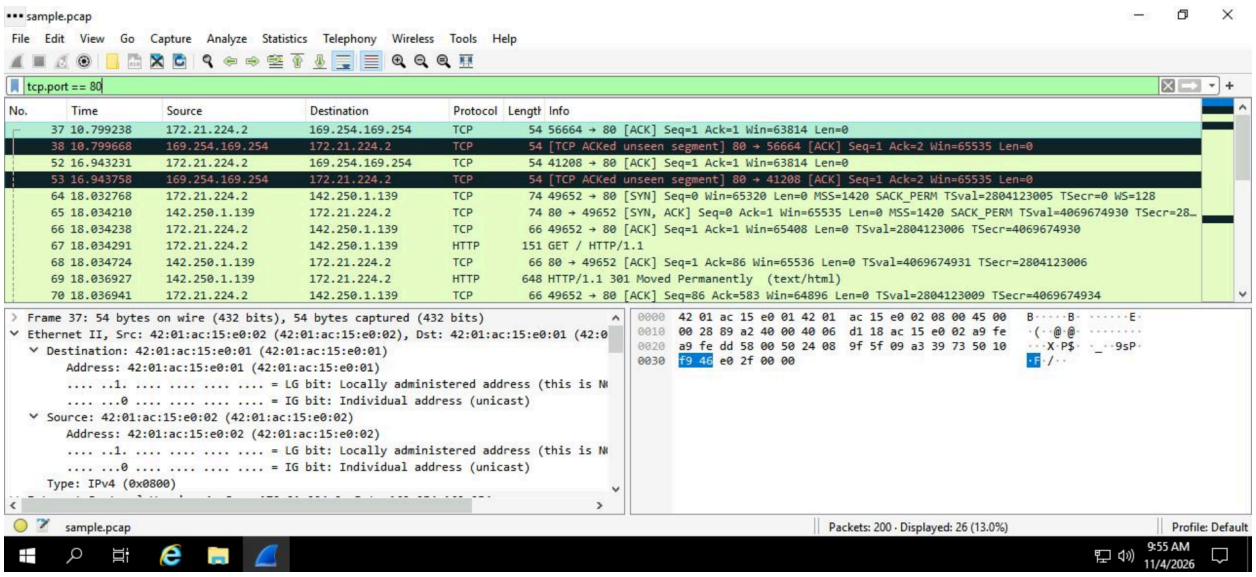
In this task, I used additional filters to select and examine TCP packets. I learnt how to search for text that is present in payload data contained inside network packets. This locates packets based on something such as a name or some other text that is of interest

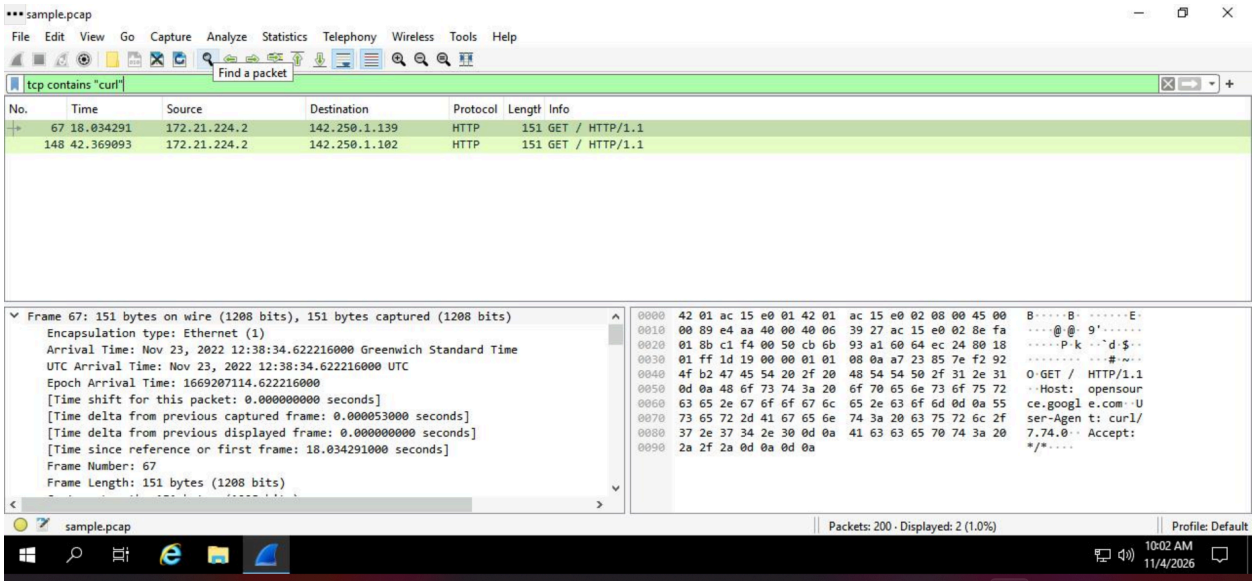
1. Enter the following filter to select TCP port **80** traffic. TCP port **80** is the default port that is associated with web traffic:
2. Click the **X** icon to close the detailed packet inspection window.
3. Click the **X Clear display filter** icon in the Wireshark filter bar to clear the filter.
4. Enter the following filter to select TCP packet data that contains specific text data.

```
tcp contains "curl"
```

Press **ENTER** or click the **Apply display filter** icon in the filter text box.

This filters to packets containing web requests made with the `curl` command in this sample packet capture file.





This is a practical experience using Wireshark to

- open saved packet capture files,
- view high-level packet data, and
- use filters to inspect detailed packet data.

This is an important milestone on my cybersecurity journey toward understanding how to use network packet analysis tools to examine network traffic!